

Clinical context

HoLEP morcellation

suprapubic view
bladder wall vs
morcellator tip

Validated here — contact-force regulation

Probe replica

3D-scanned from the
clinical GE 4C-RS
on the FR5 flange

PX6D F/T sensor

six-axis, 1 kHz
gravity and payload
compensated

Admittance loop

penetration axis only
100 Hz on $\|F\|$
10 mm/s, 0.2 rad/s

FR5 arm

6-DoF cobot
servo 125 Hz
probe placement

contact force fed back each sample

Safety supervisor — independent of image and learning

warning 4.5 N · hard limit 5.0 N with forced retreat · command watchdogs

Not validated here — future image-guided and learned components

Sonologger

ultrasound video and
IMU orientation on
one host clock

Bladder-lumen

segmentation
not integrated with
the control stack

Image-guided

orientation control
not implemented
in the control stack

Learned policy

from clinician
demonstrations
none trained

to the arm,
alongside the
force loop